



Brief paper

Observability is sufficient for the design of globally exponentially stable state observers for state-affine nonlinear systems[☆]

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ABSTRACT

In this paper we are interested in the problem of state observation of state-affine nonlinear systems. It is well-known that *observability* of the system is a *necessary* assumption for the state reconstruction. Our main contribution is to prove that it is also *sufficient* to design a *globally exponentially stable* observer. This should be contrasted with existing results that require the *strictly stronger* assumption of uniform complete observability of the system. To the best of the authors' knowledge this is the first time such a result is reported in the literature.

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1. Introduction

State estimation of dynamical systems using measurable information is a fundamental problem in the field of systems and control, with significant applications in many practical scenarios for decision-making, monitoring and control—with several comprehensive books devoted to this topic (Bernard, 2019; Besancon, 2007; Gauthier & Kupka, 2001). A classical solution is to design a dynamical observer driven by the measured output, yielding an (asymptotic) estimation of the unavailable system state. In the last decades, numerous results have been reported, the interested reader is referred to Bernard, Andrieu, and Astolfi (2022) for a thorough survey of recent developments.

In this paper, we are interested in state observer design of state-affine nonlinear systems, which have been extensively studied in the control literature. The standard solution to this problem is to view the system as a linear time-varying (LTV) system and apply a classical Kalman–Bucy filter to it. The Kalman–Bucy filter generates an error dynamics for the systems state whose zero equilibrium is *uniformly globally exponentially stable* (UGES),

provided the corresponding LTV system satisfies the assumption of *uniform complete observability* (UCO). See Definition 1 for the definition of this property and Bernard (2019, Theorem 3.3) for a precise formulation of this result, given in terms of the (backward) observability Gramian of the system. It is widely recognized that the UCO assumption is a *very restrictive* condition in many practical applications. In view of this, many researchers have proposed alternative observer designs that achieve only *global exponential stability* (GES) – without the uniform qualifier – under weaker assumptions.¹ See Aranovskiy et al. (2020), Bobtsov, Ortega, Yi, and Nikolayev (2021), Ortega, Bobtsov, Nikolayev, Schiffer, and Dochain (2021), Rapaport and Dochain (2019), Wang, Efimov, Aranovskiy, and Bobtsov (2020) for some recent contributions.

As shown in Lemma 1 a *necessary* condition to reconstruct the state of a dynamical system is that it is *observable*—a property that, for the class of systems considered in the paper, is equivalent to the fact that the set of unobservable initial conditions for the state is empty. The main contribution of the paper is the proof that this assumption is also *sufficient* to design a *GES observer* for a class of single-input single-output, state-affine nonlinear systems.

The design of the proposed observer proceeds along the following steps.

S1 We invoke the generalized parameter estimation-based observer (GPEBO) design technique reported in Ortega, Bobtsov, et al. (2021) to translate the state observation problem into a

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¹ See Remark 1 for a discussion of the distinction between UGES and GES.

problem of *parameter estimation* of a linear regressor equation (LRE).

S2 To reconstruct the parameters of the LRE we apply the parameter estimator proposed in Wang, Ortega, Bobtsov, Romero, and Yi (2021), where it is used to design new model reference adaptive controllers. The novelty of this estimator is that it invokes again the GPEBO technique to generate a new *extended* LRE to which we apply the mixing step of dynamic regressor extension and mixing (DREM) parameter estimators to generate a set of scalar LREs.² In view of its GPEBO+DREM construction procedure this estimator is called G+D.

S3 In Wang et al. (2021) it is shown that the G+D estimator ensures global exponential convergence under the weak assumption of *interval excitation* (IE) of the regressor (Kreisselmeier & Rietze-Augst, 1990; Tao, 2003). Moreover, it is shown to be equivalent to *identifiability* of the LRE, that is a necessary condition to reconstruct the parameters of a LRE. In this last step, we show that – in the context of state estimation – this condition is *equivalent* to observability of the state-affine system.

The interested readers are referred to the aforementioned papers for further details on GPEBO, DREM and the G+D design. Since the construction of the observer involves the application of GPEBO twice and then the use of DREM, we refer in the sequel to it as *2G+D observer*.

The remainder of the paper is organized as follows. In Section 2 we define the class of systems considered in the paper and give the definitions of observability and UCO. In Section 3 we complete **S1** by deriving the LRE. Section 4 recalls the main result of Wang et al. (2021), that is, the G+D parameter estimator for identifiable LRE. In Section 5 we establish the key result that observability of the system is equivalent to identifiability of the associated LRE. In Section 6 we present our main result by concluding the previous three sections corresponding to the steps **S1–S3**. An illustrative example of a non-holonomic vehicle is given to show the effectiveness of the proposed approach in Section 7. We wrap up the paper with concluding remarks and some possible extensions of the main result in Section 8.

Notation. I_n is the $n \times n$ identity matrix. For $x \in \mathbb{R}^n$, we denote the Euclidean norm $|x|^2 := x^\top x$. For a matrix $A \in \mathbb{R}^{n \times n}$, $A^\top$ denotes its transpose, $\text{tr}(A)$ is its trace, $\text{vec}(A)$ denotes its vectorization, $\text{adj}(A)$ denotes its adjoint matrix and $\|A\|$ denotes its induced matrix 2-norm. The Kronecker product between two matrices A and B is denoted by $A \otimes B$. All mappings are assumed smooth and all dynamical systems are assumed to be forward complete. Given a number $n \in \mathbb{Z}_{>0}$ we define the set $\bar{n} := \{1, 2, \dots, n\}$.

2. Problem statement

Consider the problem of state observation of single-input single-output, *state-affine* nonlinear systems with dynamics of the form

$$\begin{aligned} \dot{x}(t) &= \mathbf{A}(u, y, t)x(t) + \mathbf{b}(u, y, t), \quad x(t_0) = x_0 \in \mathbb{R}^n \\ y(t) &= \mathbf{C}(u, t)x(t), \end{aligned} \quad (1)$$

with $x(t) \in \mathbb{R}^n$, $y(t) \in \mathbb{R}$, $u(t) \in \mathbb{R}$ and initial time $t_0 \geq 0$. Following the standard procedure we write the system (1) as an LTV system

$$\dot{x} = A(t)x + b(t), \quad x(t_0) = x_0$$

² In classical DREM designs the extended regressor is generated applying LTV operators to the original LRE, see Aranovskiy, Bobtsov, Ortega, and Pyrkin (2017), Ortega, Aranovskiy, Pyrkin, Astolfi, and Bobtsov (2021) for further details on DREM.

$$y = C(t)x \quad (2)$$

where we defined $A(t) := \mathbf{A}(u(t), y(t), t)$, $b(t) := \mathbf{b}(u(t), y(t), t)$, and $C(t) := \mathbf{C}(u(t), t)$. Note that, as $\mathbf{A}$, $\mathbf{b}$ are functions of the output y , the matrices A , b depend on the initial conditions (t_0, x_0) .³

The observability Gramian of the system (2) is defined as

$$W(t_0, t_1) := \int_{t_0}^{t_1} \Phi^\top(\tau, t_0) C^\top(\tau) C(\tau) \Phi(\tau, t_0) d\tau \quad (3)$$

where $t_1 \geq t_0 \geq 0$ and $\Phi(t, t_0)$ is the state transition matrix of the homogeneous system $\dot{x}(t) = A(t)x(t)$.

The following result is well-known in observer theory.

Lemma 1. (Rugh, 1996, Theorem 9.8) Consider the system (2). The following statements are equivalent.

- There exists $T > 0$ such that $W(t_0, t_0 + T)$ is invertible.
- The system is observable in the interval $\mathcal{I} := [t_0, t_0 + T]$, that is, the initial condition x_0 can be uniquely determined from measurements of y and u in $\mathcal{I}$.
- The set of unobservable initial conditions in $\mathcal{I}$, that is,

$$\{x_0 \in \mathbb{R}^n / \{0\} \mid C(t)\Phi(t, t_0)x_0 = 0, \quad \forall t \in \mathcal{I}\}$$

is the empty set.

In view of the above result, it is clear that if the system (2) is unobservable, then there exist initial conditions x_0 that cannot be reconstructed from $y(t)$, $u(t)$. This in turn demonstrates the necessity of the observability to reconstruct the system state of (2). On the other hand, as indicated in the Introduction, it is well-known that a classical Kalman–Bucy filter ensures UGES of the estimated state if the system (2) is UCO, a property that is defined as follows.

Definition 1. (Rugh, 1996) The system (2) with bounded A , C is UCO if

$$W(t, t + T) \geq \delta I_n, \quad \forall t \geq t_0, \quad (4)$$

for some $T > 0$ and $\delta > 0$.

The requirement that the constants T and δ are *independent* of t_0 makes the UCO property very restrictive, but it endows the UGES property of the Kalman–Bucy filter with the “uniform” qualifier.

Problem Formulation Assume the system (1) (equivalently, (2)) is *observable*. Design an observer of the form

$$\begin{aligned} \dot{\hat{x}} &= F(\chi, y, t) \\ \hat{x} &= H(\chi, y, t), \end{aligned} \quad (5)$$

$\chi(t) \in \mathbb{R}^{n_\chi}$, such that the dynamics of the state estimation error $\tilde{x} := \hat{x} - x$, satisfies the following properties. First, its zero equilibrium is Lyapunov stable. Second, for every $\rho_x > 0$ there exists $m(t_0, \rho_x) > 0$ and $\lambda(t_0, \rho_x) > 0$, such that if $|\tilde{x}(t_0)| \leq \rho_x$ then the associated trajectory $\tilde{x}(t)$ satisfies the bound

$$|\tilde{x}(t)| \leq m e^{-\lambda(t-t_0)} |\tilde{x}(t_0)|, \quad \forall t \geq t_0. \quad (6)$$

That is, the zero solution of the observation error dynamics is GES.

Following standard practice in observer theory (Bernard, 2019), we assume that the state trajectories of (2) are bounded. Moreover, we assume that the state transition matrix of the homogeneous part of the system (2) verifies

$$\|\Phi(t, \tau)\| \leq \phi_M, \quad \forall t \geq \tau \geq t_0, \quad x_0 \in \mathbb{R}^n. \quad (7)$$

³ See Theorem 3.3, and the discussion above, of Bernard (2019) for a precise statement concerning this key issue. As will be shown below, our stability claims for both cases will be different, consequently, we will distinguish between state-affine or LTV system whenever needed.

It is well-known (Rugh, 1996, Theorem 6.4) that this assumption is equivalent to *uniform stability* of the homogeneous part of the system (2)—consequently, in the case of LTV systems, the number ϕ_M is *independent* of (t_0, x_0) . This assumption may be relaxed incorporating an output injection to achieve the latter objective—but we skip this step for brevity.

Remark 1. The above GES notion is a simple generalization of the exponential stability in Jenkins, Annaswamy, Lavretsky, and Gibson (2018, Definition 2(i)) by including the possible dependence of m, λ in (6) on t_0 .⁴ Note that since the convergence parameters m and λ may depend on $(t_0, \tilde{x}(t_0))$, the GES property is *not uniform*. It is widely known that the uniformity property with m, λ in (6) independent on $(t_0, \tilde{x}(t_0))$ endows the system with stronger robustness properties with respect to external perturbations than simple GES.

3. Problem transformation to parameter estimation

In this section, we complete the step S1, that is, following the GPEBO approach we reformulate the problem of state observation as one of parameter estimation. Towards this end, we derive a LRE that is going to be used for the parameter estimation. Although the result is a particular case of Ortega, Bobtsov, et al. (2021, Proposition 1), which is applicable to a broader class of nonlinear systems and includes a coordinate change, for the sake of completeness we give the result in detail and include its proof.

Lemma 2. Consider the system (2) and the GPEBO dynamics

$$\dot{\xi} = A(t)\xi + b(t), \quad \xi(t_0) = 0 \quad (8a)$$

$$\dot{\Phi}_A = A(t)\Phi_A, \quad \Phi_A(t_0) = I_n. \quad (8b)$$

The resulting signals $\xi(t)$ and $\Phi_A(t)$ are bounded, and the state of the system (2) verifies the relation

$$x = \xi + \Phi_A \theta, \quad (9)$$

where the unknown constant vector $\theta \in \mathbb{R}^n$ satisfies the LRE

$$\mathcal{Y} = \psi^\top \theta, \quad (10)$$

and the measurable signals $\mathcal{Y}(t) \in \mathbb{R}$ and $\psi(t) \in \mathbb{R}^n$ are defined by

$$\mathcal{Y} := y - C(t)\xi \quad (11a)$$

$$\psi := \Phi_A^\top C^\top(t). \quad (11b)$$

Proof. The error signal $e := x - \xi$ satisfies the LTV dynamics $\dot{e} = A(t)e$. Now, from (8b) we see that Φ_A is the fundamental matrix⁵ of the e system, which satisfies (Rugh, 1996, Property 4.4)

$$\Phi_A(t) = \Phi(t, t_0), \quad (12)$$

with Φ defined after (3). Consequently, there exists a constant vector $\theta \in \mathbb{R}^n$ such that $e = \Phi_A \theta$, namely $\theta = e(t_0) = x_0$. We now have the following chain of implications

$$\begin{aligned} e = \Phi_A \theta &\Leftrightarrow x = \xi + \Phi_A \theta \quad (\Leftarrow e = x - \xi) \\ &\Rightarrow C(t)x = C(t)\xi + C(t)\Phi_A \theta \\ &\Leftrightarrow y - C(t)\xi = \psi^\top \theta \quad (\Leftarrow (2), (11b)) \\ &\Leftrightarrow \mathcal{Y} = \psi^\top \theta \quad (\Leftarrow (11a)). \end{aligned}$$

The claim of boundedness of the signals ξ, Φ_A follows from the assumption (7) and boundedness of x . $\square$

⁴ In Jenkins et al. (2018, Definition 2(i)) the qualifier *global* is omitted, but it is implicit in the definition. Besides, it is uniform in t_0 , implying uniform asymptotic stability.

⁵ Also called principal matrix.

4. Parameter estimation of (10) via G+D

In the previous section, the state observation problem of system (2) has been transformed into the parameter estimation problem of the LRE (10). In this section we complete the step S2 by recalling the main result of Wang et al. (2021), namely that identifiability of the LRE (10) is sufficient to design a globally exponentially stable parameter estimator for the LRE (10).

4.1. Identifiability is equivalent to IE

To streamline the presentation of the results we introduce the following definition and recall a key lemma from Wang et al. (2021, Proposition 1), whose proof is omitted for brevity.

Definition 2. The LRE (10) is *identifiable* if and only if there exists a set of strictly increasing time instants $\{t_i\}_{i \in \bar{n}}$ such that

$$\text{rank} \left\{ \begin{bmatrix} \psi(t_1) & \psi(t_2) & \cdots & \psi(t_n) \end{bmatrix} \right\} = n.$$

Lemma 3. The LRE (10) is identifiable if and only if the regressor ψ is IE, i.e.,

$$\int_{t_0}^{t_c} \psi(s) \psi^\top(s) ds \geq c_c I_n, \quad (13)$$

for some $c_c > 0$ and $t_c > t_0$.⁶

4.2. The G+D parameter estimator

Bearing in mind Lemma 3, we now present the main result of the section.

Lemma 4. Consider the LRE (10). Define the G+D interlaced estimator

$$\dot{\hat{\theta}}_g = \gamma_g \psi(\mathcal{Y} - \psi^\top \hat{\theta}_g), \quad \hat{\theta}_g(t_0) = 0 \quad (14a)$$

$$\dot{\hat{\Omega}} = -\gamma_g \psi \psi^\top \hat{\Omega}, \quad \hat{\Omega}(t_0) = I_n \quad (14b)$$

$$\dot{\hat{\theta}} = \Gamma \Delta(Y - \Delta \hat{\theta}), \quad \hat{\theta}(t_0) = 0, \quad (14c)$$

where $\gamma_g > 0$, $\Gamma = \text{diag}\{\gamma_1, \dots, \gamma_n\} > 0$, and we define

$$\Delta := \det\{I_n - \hat{\Omega}\} \quad (15a)$$

$$Y := \text{adj}\{I_n - \hat{\Omega}\} \hat{\theta}_g. \quad (15b)$$

If the LRE (10) is identifiable, then for all $\theta \in \mathbb{R}^n$, we have the following.

- (i) The error dynamics for each of the elements the parameter estimation error $\tilde{\theta}_i := \hat{\theta}_i - \theta_i$, $i \in \bar{n}$, is given by

$$\dot{\tilde{\theta}}_i = -\gamma_i \Delta^2 \tilde{\theta}_i, \quad (16)$$

whose zero equilibrium is Lyapunov stable.

- (ii) For any $\rho > 0$, there exist $m_p(t_0, \rho) > 0$ and $\lambda_p(t_0, \rho) > 0$, such that if $|\tilde{\theta}(t_0)| \leq \rho$ then the associated trajectory $\tilde{\theta}(t)$ satisfies the bound

$$|\tilde{\theta}(t)| \leq m_p \exp(-\lambda_p(t - t_0)) |\tilde{\theta}(t_0)|, \quad \forall t \geq t_0.$$

- (iii) All signals $\hat{\theta}_g(t), \hat{\Omega}(t), \hat{\theta}(t)$ are bounded.

⁶ We recall (Sastrey & Bodson, 1989) that ψ is PE if there exists $T > 0$ and $\delta_c > 0$ such that $\int_t^{t+T} \psi(s) \psi^\top(s) ds \geq \delta_c I_n$, $\forall t \geq t_0$.

Proof. Replacing (10) in (14a) yields the error dynamics for the gradient estimator

$$\dot{\tilde{\theta}}_g = -\gamma_g \psi \psi^\top \tilde{\theta}_g, \quad (17)$$

where $\tilde{\theta}_g := \hat{\theta}_g - \theta$. From (17) and the properties of the *fundamental matrix* Ω , defined in (14b), we get $\tilde{\theta}_g = \Omega \tilde{\theta}_g(t_0)$, which may be rewritten as the extended LRE

$$(I_n - \Omega)\theta = \hat{\theta}_g. \quad (18)$$

Multiplying (18) by $\text{adj}\{I_n - \Omega\}$ we get the following new LRE

$$Y = \Delta\theta, \quad (19)$$

with Δ and Y defined in (15a) and (15b), respectively. We underscore the fact that the regressor Δ is a *scalar*. Replacing (19) in (14c) yields the error dynamics (16) for each $\tilde{\theta}_i$, proving the claim (i).

Now, the explicit solution of (16) is given by

$$\tilde{\theta}_i(t) = \exp\left(-\gamma_i \int_{t_0}^t \Delta^2(s) ds\right) \tilde{\theta}_i(t_0).$$

In view of Lemma 3, the proof is completed if the statements (ii) and (iii) are proved under IE of ψ . Towards this end, it is shown in Lemma 5 in Appendix A (see also Wang et al. (2021, Lemma 3)) that if ψ is IE, $|\Delta(t)| \geq \epsilon$, for all $t \geq t_c$, where ϵ is defined in (A.1). Thus, we have

$$|\tilde{\theta}_i(t)| \leq m_{p,i} \exp(-\gamma_i \epsilon^2 (t - t_0)) |\tilde{\theta}_i(t_0)| \quad (20)$$

with $m_{p,i} := e^{\gamma_i \epsilon^2 (t_c - t_0)}$. This proves the statement (ii) with $m_p := \max_{i \in \bar{n}} \{m_{p,i}\}$ and $\lambda_p := \epsilon^2 \min_{i \in \bar{n}} \{\gamma_i\}$. In view of (A.1), we see that the convergence parameters m_p and λ_p depend on the IE parameters (t_c, c_c) in (13) and ψ_M given in (A.2).

To prove the claim (iii), we define the functions

$$V_{\tilde{\theta}_g}(\tilde{\theta}_g) := \frac{1}{2} |\tilde{\theta}_g|^2, \quad V_\Omega(\Omega) := \frac{1}{2} \text{tr}(\Omega^\top \Omega). \quad (21)$$

Taking their time derivatives yields⁷

$$\begin{aligned} \dot{V}_{\tilde{\theta}_g} &= -|\psi \tilde{\theta}_g|^2 \leq 0 \\ \dot{V}_\Omega &= -\gamma_g \text{vec}(\Omega)^\top \text{blockdiag}\{\psi \psi^\top, \dots, \psi \psi^\top\} \text{vec}(\Omega) \leq 0. \end{aligned}$$

This implies that $\tilde{\theta}_g(t)$ and $\Omega(t)$ are both bounded. As a result, recalling the statement (ii), we conclude that all signals $\tilde{\theta}_g(t)$, $\Omega(t)$, $\hat{\theta}(t)$ are bounded. $\square$

Remark 2. Recalling (16) and Lemma 5 that ensures $|\Delta(t)| \geq \epsilon$ for $t \geq t_c$, it follows that the exponential convergence rate of the parameter estimation error $\tilde{\theta}$ is tunable via the parameters γ_i , i.e., a larger γ_i leads to a faster convergence speed of $\hat{\theta}_i(t)$ to the actual value θ_i .

Remark 3. It is important to make the following clarification regarding the *identification mechanism* of the proposed observer. If at some time before the estimate has had time to converge the regressor ψ in the original LRE $\mathcal{Y} = \psi^\top \theta$ lacks PE, there is no new information in this LRE that can be used to bring the estimate to the right value. However, notice that DREM uses the new LRE (19), i.e., $Y = \Delta\theta$ where the past information of ψ is gathered in Δ via its integration. This is clear recalling the definition of Δ given in (14b) and (15a). A key point here is that Δ is a *scalar* and the new LRE can be written component-wise as $Y_i = \Delta\theta_i$, $i = 1, \dots, q$. Once the LRE (10) is identifiable, or

⁷ We use the relations $\text{vec}(ABC) = (C^\top \otimes A)\text{vec}(B)$ and $\text{tr}(AC) = \text{vec}(A^\top)^\top \text{vec}(C)$, where A, B and C denote matrices of appropriate dimensions.

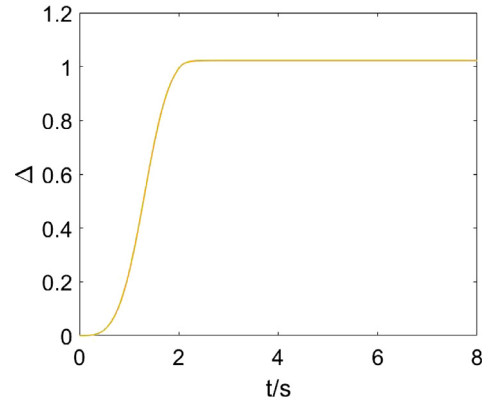


Fig. 1. Signal Δ .

equivalently the regressor ψ is IE, it has been shown in Lemma 5 in Appendix A that $|\Delta(t)| \geq \epsilon$ for some $\epsilon > 0$ and all $t \geq t_c$. This ensures that, being a scalar, the regressor Δ is PE, guaranteeing the exponential convergence of the estimate to the right value.

4.3. An example

To illustrate the effectiveness of the proposed G+D interlaced estimator 4, we consider the LRE (10) with

$$\psi(t) = \begin{cases} \begin{bmatrix} 2-t \\ 2t-1 \end{bmatrix}, & t \leq 2 \\ \begin{bmatrix} 0 \\ 3 \end{bmatrix}, & t > 2 \end{cases}, \quad \theta = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Let the initial time $t_0 = 0$. It is clear that the regressor vector $\psi(t)$ is identifiable and IE, but not PE. We apply the G+D interlaced estimator 4 to estimate the unknown parameter θ . Let $\gamma_g = 1$. In Fig. 1, the simulated trajectory of $\Delta(t)$ is presented, showing that $\Delta(t)$ starts from zero and gradually grows to a positive constant—validating the PE of Δ .

To illustrate the claim of Remark 2, the trajectories of the estimated parameters $\hat{\theta} = \text{col}(\hat{\theta}_1, \hat{\theta}_2)$ are given in Fig. 2 for $\Gamma = \gamma I_2$ with three different adaptation gains $\gamma = 1, 2, 4$. It is seen that the speed of convergence of $\hat{\theta}_i$ increases with larger gains. On the other hand, it is noted that by letting $t_0 \geq 2$ the IE of ψ is not satisfied. More explicitly, the first entry of θ does not appear in the LRE, and thus it is impossible to design an asymptotic estimator of θ . This demonstrates that the established exponential stability of the equilibrium $\tilde{\theta} = 0$ is not uniform.

5. Equivalence between observability of (2) and identifiability of (10)

In this section we prove that observability of the LTV system (2) is *equivalent* to identifiability of the LRE (10), completing the step S3.

Proposition 1. *The following statements are equivalent.*

- (i) *The LTV system (2) is observable.*
- (ii) *The LRE (10) is identifiable.*

Proof. (i) $\Rightarrow$ (ii). We prove (i) $\Rightarrow$ (ii) by contradiction. We suppose that the LRE (10) is *not* identifiable, i.e., there exists a positive integer $n_0 < n$ such that

$$\text{rank}[\psi(t_1) \dots \psi(t_n)] \leq n_0 \quad (22)$$

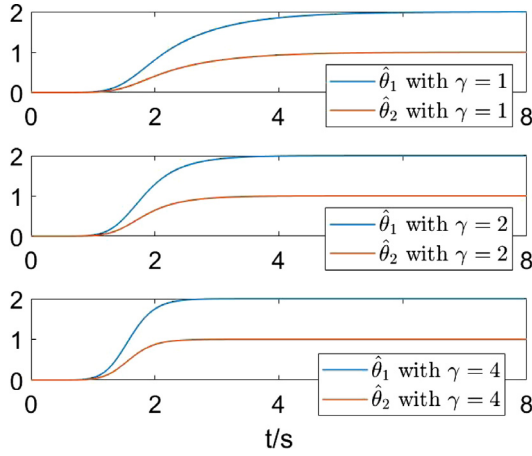


Fig. 2. Estimated parameters $\hat{\theta}_1$ and $\hat{\theta}_2$ for three different adaptation gains.

holds for all time sequence $\{t_i\}_{i \in \bar{n}}$, with $\{\bar{t}_i\}_{i \in \bar{n}}$ being such that $\text{rank}[\psi(\bar{t}_1) \dots \psi(\bar{t}_n)] = n_0$.

Let $v \in \mathbb{R}^n$ be such that $|v| = 1$ and

$$\psi^\top(\bar{t}_i)v = 0, \quad \forall i \in \bar{n}.$$

Next, we show that $\psi^\top(t)v = 0$ for all $t \in \mathbb{R}_{\geq 0}$ by contradiction. We suppose that there exists a $t^\# \in \mathbb{R}_{\geq 0}$ such that $\psi^\top(t^\#)v \neq 0$. This indicates that

$$n_0 = \text{rank}[\psi(\bar{t}_1) \dots \psi(\bar{t}_n)] < \text{rank}[\psi(\bar{t}_1) \dots \psi(\bar{t}_n) \psi(t^\#)]$$

which contradicts with the assumption that (22) holds for all time sequence $\{t_i\}_{i \in \bar{n}}$. Hence, we have $\psi^\top(t)v = 0$ for all $t \in \mathbb{R}_{\geq 0}$.

With this in mind, defining $z(t) := \Phi_A(t)v$, and invoking (8b), we get $\dot{z}(t) = A(t)z(t)$, which yields

$$\begin{aligned} v^\top W(t_0, t_f)v &= \int_{t_0}^{t_f} |C(t)\Phi_A(t)v|^2 dt \\ &= \int_{t_0}^{t_f} |\psi^\top(t)v|^2 dt = 0 \end{aligned}$$

for all $t_f \geq t_0$. This clearly contradicts with the observability of the LTV system (2). Therefore, there exists a time sequence $\{t_i\}_{i \in \bar{n}}$ such that $\text{rank}[\psi(t_1) \dots \psi(t_n)] = n$, proving the statement (ii).(ii) $\Rightarrow$ (i). To prove the statement (i) we let $T > t_n$, and proceed to show that

$$W(t_0, T) := \int_{t_0}^{t_0+T} \Phi_A^\top(t)C^\top(t)C(t)\Phi_A(t)dt > 0.$$

From the identifiability assumption we have that the matrix $[\psi(t_1) \dots \psi(t_n)]$ is full rank, it can be seen that for any $v \in \mathbb{R}^n$ satisfying $|v| = 1$, there always exists a $\bar{i} \in \bar{n}$ such that $|\psi(\bar{t}_i)v| > 0$. By continuity, it follows that for any $v \in \mathbb{R}^n$ satisfying $|v| = 1$, there exists an $\epsilon > 0$ such that

$$\sum_{i \in \bar{n}} |\psi(\bar{t}_i)v| \geq |\psi(\bar{t}_{\bar{i}})v| > 0, \quad \forall \hat{t}_i \in [t_i, t_i + \epsilon],$$

yielding

$$\begin{aligned} v^\top W(t_0, t_0 + T)v &= \int_{t_0}^{t_0+T} |C(t)\Phi_A(t)v|^2 dt \\ &\geq \int_{\bar{t}_{\bar{i}}}^{t_0+T} |\psi^\top(t)v|^2 dt > 0. \end{aligned}$$

Therefore, by recalling that such v is arbitrary, it can be concluded that $W(t_0, t_0 + T)$ is nonsingular for $t_0 + T > t_n$, proving the observability of the LTV system (2). $\square$

6. Main result

In view of the previous three sections, we now give our main result in the proposition below.

Proposition 2. The 2G+D observer proposed in Lemmata 2 and 4 provides a solution to the problem formulated in Section 2 with a 2G+D observer of the form (5) with $n_x = 3n + 2n^2$.⁸

Proof. Let the 2G+D observer (5) be a cascade of the GPEBO dynamics 2 and the G+D interlaced estimator 4, and the observed state

$$\hat{x} = \xi + \Phi_A \hat{\theta}. \quad (23)$$

Under the observability assumption of system (2), it immediately follows that for all initial conditions $x(t_0) \in \mathbb{R}^n$, the parameter estimation error $\tilde{\theta}(t)$ globally exponentially converges to zero and all signals $(\xi, \Phi_A, \theta_g, \Omega, \hat{\theta})$ are bounded by recalling Lemmas 2, 4 and Proposition 1. Further, with (7) and (9) we have

$$|\tilde{x}(t)| \leq \|\Phi_A\| |\tilde{\theta}(t)| \leq \phi_M |\tilde{\theta}(t)|, \quad (24)$$

which, together with $\tilde{x}(t_0) = \tilde{\theta}(t_0)$ and Lemma 4.(i) and (ii), completes the proof. $\square$

Remark 4. By (24) and Lemma 4.(ii), the corresponding convergence parameters (m, λ) in (6) depend on ϕ_M in (7), and λ_p, m_p . In the proof of Lemma 4 we showed that the latter depend on the IE parameters (t_c, c_c) in (13) and ψ_M given in (A.2). From (7) and the definition of ψ in (11b) we conclude that, for LTV systems, m_p and λ_p depend only on t_c and c_c , hence on t_0 . On the other hand, for state affine systems, when Φ_A and C may depend on x_0 , they are functions of both t_0 and x_0 . In any case, as discussed in Remark 1, the GES property that is established in the paper is *not uniform*.

7. State estimation of non-holonomic vehicles

Consider the non-holonomic vehicle of the form (Bernard, 2019)

$$\begin{aligned} \dot{z}_1 &= u_1 \cos(z_3) \\ \dot{z}_2 &= u_1 \sin(z_3) \\ \dot{z}_3 &= u_1 u_2 \end{aligned} \quad (25)$$

where the inputs u_1 and u_2 represent the norm of the vehicle velocity and the orientation of the front steering wheels, respectively, and the measurable outputs are $y = \text{col}(z_1, z_2)$. We follow the linearization approach of Bernard (2019, Subsection 6.3.3) by defining

$$x_1 = z_1, \quad x_2 = z_2, \quad x_3 = \cos(z_3), \quad x_4 = \sin(z_3).$$

In this way, the system (25) is transformed into the form (1) that, with the definitions

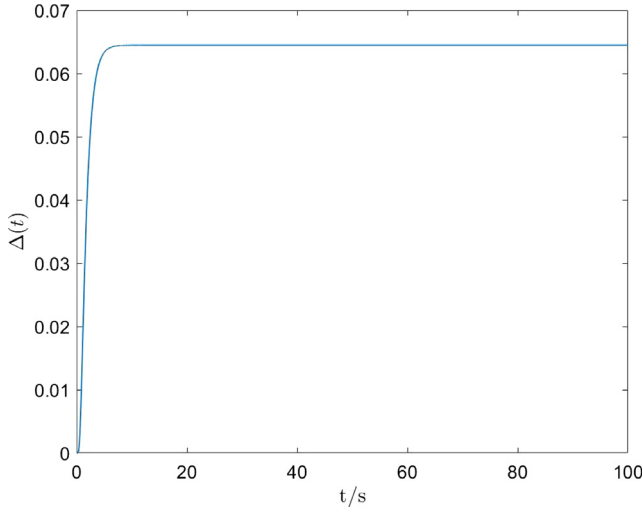
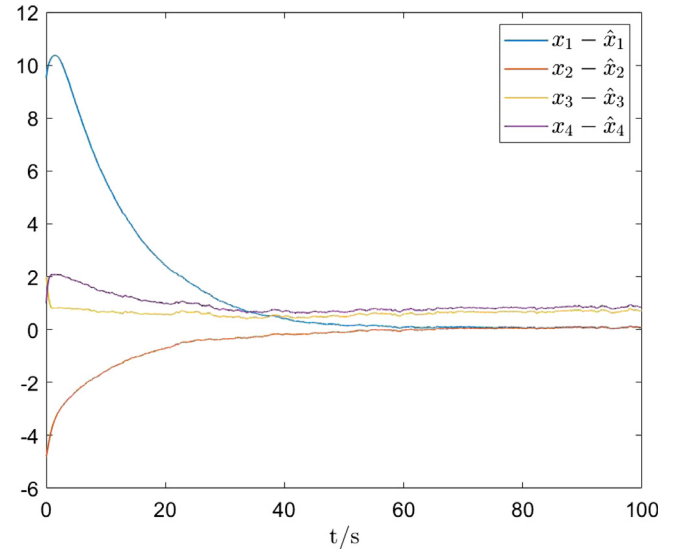
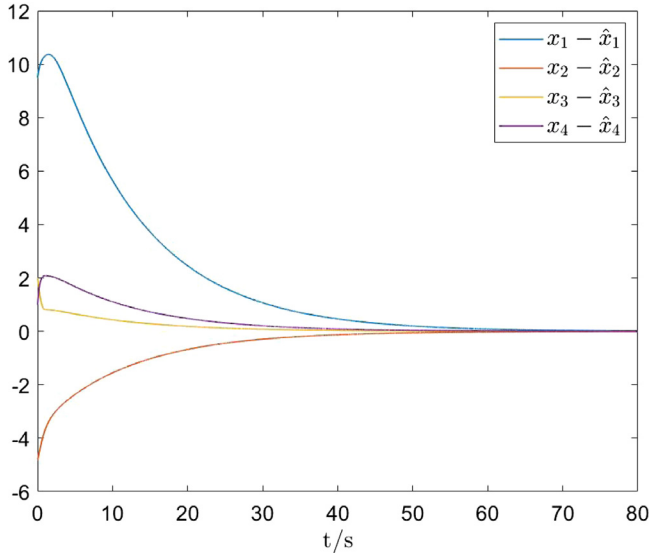
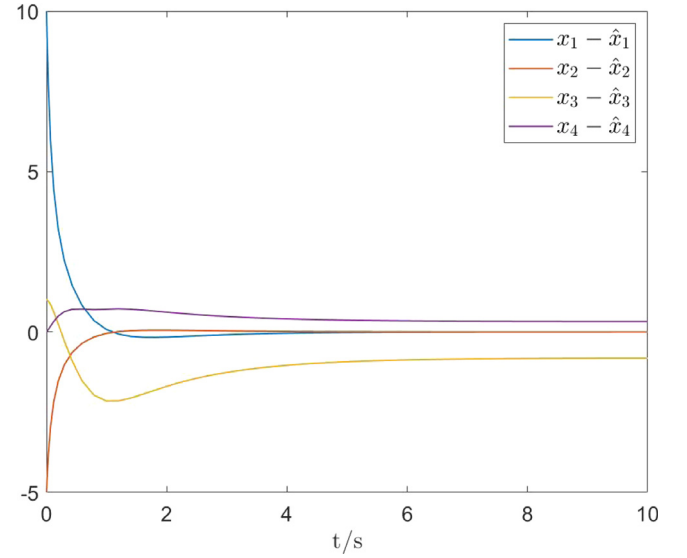
$$A(t) = \begin{bmatrix} 0 & 0 & u_1(t) & 0 \\ 0 & 0 & 0 & u_1(t) \\ 0 & 0 & 0 & -u_1(t)u_2(t) \\ 0 & 0 & u_1(t)u_2(t) & 0 \end{bmatrix}, \quad C^\top = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix},$$

can be compactly written as (2) with $b(t) = 0$. Let

$$u_1 = e^{-t}, \quad u_2 = \begin{cases} 2 - 2t, & t \leq 1 \\ 0, & t > 1. \end{cases}$$

It then follows that the resulting LTV system is observable but *not UCO*. We apply the proposed 2G+D observer 2 and 4 to estimate

⁸ It is worth pointing out that the design of a Kalman–Bucy filter requires the solution of a Riccati differential equation of dimension $\frac{1}{2}n(n+1)$.

Fig. 3. Signal $\Delta(t)$.Fig. 5. State estimation errors $x - \hat{x}$ under measurement noises.Fig. 4. State estimation errors $x - \hat{x}$.Fig. 6. State estimation errors $x - \hat{x}$ using the Kalman-Bucy's filter of Bernard (2019, Theorem 3.3).

the state $x(t)$, and the resulting simulations using $\gamma_g = 10$ and $\Gamma = 20I_4$ are given in Figs. 3 and 4. It can be seen from Fig. 3 that the signal $\Delta(t)$ is strictly larger than 0.06 after 5 seconds and from Fig. 4 that the observed states $\hat{x}(t)$ asymptotically converge to the actual states $x(t)$.

To study the robustness of the proposed observer, we consider the case with measurement noises which are generated by the block “Uniform Random Number” of Simulink. The resulting simulation results are presented in Fig. 5, where small state estimation errors are observed, showing robustness to measurement noise.

As a comparison, we also design the Kalman-Bucy's filter of Bernard (2019, Theorem 3.3) to derive a state estimate. The simulation result is given in Fig. 6, where it is seen that the state estimation errors $x_3 - \hat{x}_3$ and $x_4 - \hat{x}_4$ are not converging to zero. This fact is not surprising since there is no guarantee of asymptotic state estimation using the Kalman-Bucy's filter for systems that are not UCO.

8. Concluding remarks and some extensions

We have presented in the paper a 2G+D state observer for the system (2) that ensures GES of the observation error dynamics for observable systems. The observer, consisting of the dynamic equations 2 and 4 with the auxiliary signals 2 and 4, is clearly of the form (5). The observed state is computed via (23). To the best of the authors' knowledge this is the first time a result of this level of generality – concerning state-affine nonlinear systems – is reported in the literature.

It is clear from (19) that it is possible to implement an observer with *finite convergence time*, that would generate the right estimate after t_c . Unfortunately, since we do not have a lower bound for the IE level c_c , this would be a fragile operation.

The parameter estimation method in Section 4 lies in utilizing the historical information by integrating both sides of the LRE (10) and establishing a new LRE (19) for parameter estimation. By doing so, if there exists an additive disturbance to the LRE (10), i.e., $y = \psi^\top \theta + d$, the effect of the disturbance turns out

non-vanishing, and we cannot conclude asymptotic parameter estimation even if the disturbance is vanishing. Moreover, the integration of the (bounded) disturbance may be unbounded. Despite of this, the proposed 2G+D observer may be easily adapted to enjoy some *robustness* properties, inherited from the robustness of the G+D parameter estimator of Section 3. Indeed, it is shown in Wang et al. (2021, Proposition 4) that, allowing the adaptation gain γ_g of (14a) to be time-varying, the G+D estimator defines a bounded-input-bounded-state operator with respect to additive disturbances to the LRE (10).

There are several immediate extensions of our result, in particular the case of uncertainty in the system matrices, treated in Bobtsov, Ortega, et al. (2021). Also, the 2G+D observer can be easily adapted when delay measurements are present as done in Bezzubov, Bobtsov, Efimov, Ortega, and Nikolayev (2022), Bobtsov, Nikolayev, Ortega, and Efimov (2021). In those papers the parameter estimation is carried out with a standard DREM estimator, which imposes some stronger excitation constraints. Using the G+D estimator of Wang et al. (2021) these requirements will clearly be relaxed.

Appendix A. A key technical lemma

Lemma 5. (Wang et al., 2021, Lemma 3) *Given the regressor ψ satisfying the IE condition (13), the scalar Δ defined in 4, satisfies*

$$|\Delta(t)| \geq \epsilon, \quad t \geq t_c.$$

where $\epsilon > 0$ is defined as

$$\epsilon := \left(1 - \sqrt{1 - \frac{\gamma_g c_c}{1 + \gamma_g^2 t_c^2 \psi_M^2}}\right)^n, \quad (\text{A.1})$$

with

$$\psi_M := \max_{t \in [t_0, t_c]} |\psi(t)|^2. \quad (\text{A.2})$$

Proof. In Efimov and Fradkov (2015, Lemma 1) it is shown that, for an arbitrary nonzero $\tilde{\theta}_g(t_0)$ and under the IE condition (13), the LTV system (17) satisfies the bound

$$V_{\tilde{\theta}_g}(t_0) - V_{\tilde{\theta}_g}(t_c) \geq \frac{\gamma_g c_c}{1 + \gamma_g^2 t_c^2 \psi_M^2} V_{\tilde{\theta}_g}(t_0), \quad (\text{A.3})$$

where $V_{\tilde{\theta}_g}$ is defined in (21). From (A.3), we get the bound

$$V_{\tilde{\theta}_g}(t_c) = \frac{1}{2} |\Omega(t_c) \tilde{\theta}_g(t_0)|^2 \leq \frac{1}{2} \left(1 - \frac{\gamma_g c_c}{1 + \gamma_g^2 t_c^2 \psi_M^2}\right) |\tilde{\theta}_g(t_0)|^2.$$

Since $\dot{V}_{\tilde{\theta}_g}(t) \leq 0$, it immediately follows that

$$|\Omega(t) \tilde{\theta}_g(t_0)|^2 \leq \left(1 - \frac{\gamma_g c_c}{1 + \gamma_g^2 t_c^2 \psi_M^2}\right) |\tilde{\theta}_g(t_0)|^2, \quad \forall t \geq t_c.$$

We make at this point the observation that, from (13) and the definition of ψ_M , we conclude that $\psi_M t_c \geq c_c$ and, consequently

$$\frac{\gamma_g c_c}{1 + \gamma_g^2 t_c^2 \psi_M^2} < 1. \quad (\text{A.4})$$

Now, as $\tilde{\theta}_g(t_0)$ is an arbitrary vector, it follows that the spectral radius of $\Omega(t)$ satisfies

$$\rho\{\Omega(t)\} \leq \sqrt{1 - \frac{\gamma_g c_c}{1 + \gamma_g^2 t_c^2 \psi_M^2}} < 1, \quad \forall t \geq t_c, \quad (\text{A.5})$$

where the second bound follows from (A.4). Consequently, each eigenvalue λ_i , $i \in \bar{n}$, of $I_n - \Omega(t)$ satisfies

$$|\lambda_i\{I_n - \Omega(t)\}| \geq 1 - \sqrt{1 - \frac{\gamma_g c_c}{1 + \gamma_g^2 t_c^2 \psi_M^2}} > 0, \quad \forall t \geq t_c. \quad (\text{A.6})$$

The proof is completed recalling (15a) and that the determinant is the product of the eigenvalues. $\square$

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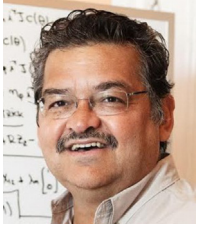
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